master can't mark this cluster update a success. In such cases, the master keeps track of these nodes and waits for a grace period set by the `cluster.follower_lag.timeout` attribute, which defaults to 90 s. If the node(s) do not respond in this 90 s grace period, they are marked as failed nodes, and the master removes them from the cluster.

As you may have gathered, a lot goes on under the hood in Elasticsearch. Cluster updates happen frequently, and the master is responsible for maintaining the moving parts. In the previous cluster updates scenario, the master awaits acknowledgments from a group of master-eligible nodes called a quorum before committing the state, rather than waiting for the rest of the nodes. A quorum is the minimum number of master nodes required for a master to operate effectively, as discussed in the next section.

14.6.4 A quorum

The master is in control of maintaining and managing the cluster. However, it consults a quorum of master-eligible nodes for cluster state updates and master elections. A quorum is a carefully selected subset of master-eligible nodes required for a master to operate the cluster effectively. This is the majority of the nodes consulted by the master to reach a consensus on matters of cluster state and other problems.

Although we are learning about a quorum, the good news is that we (users/administrators) don't have to worry about how to form one. Clusters automatically formulate a quorum based on the available master-eligible nodes. There's a simple formula to find the minimum number of master-eligible nodes (quorum) required, given a set of master-eligible nodes:

```
Minimum number of master-eligible nodes = (number of master-eligible nodes /
     2) + 1
```

Suppose we have a 20-node cluster and 8 nodes assigned as master-eligible nodes (the node role is set to `master`). By applying this formula, our cluster creates a quorum with five (carefully chosen) master-eligible nodes (8 / 2 + 1 = 5). The idea is that we need at least five master-eligible nodes to form a quorum.

The rule of thumb is that the recommended minimum number of master-eligible members in any node cluster is three. Setting three master-eligible nodes as a minimum is a surefire way to manage the cluster. Another big advantage of having at least three nodes in a cluster quorum is that this alleviates the split-brain problem, which we look at in the following section.

14.6.5 The split-brain problem

Elasticsearch's cluster health heavily relies on multiple factors: network, memory, JVM garbage collection, and so on. In some instances, the cluster is split into two clusters with some nodes in one cluster and some in the other. For example, figure 14.16 shows a cluster with two master-eligible nodes, but one (Node A) is elected to be the master node. As long as we are in a happy state, the cluster is healthy, and the master carries out its responsibilities diligently.